

Chapter 4 — Web and Media Design

Questions & Answers — Exam-Style Practice

Follows the textbook question progression: Worked Example → Try → Exam-Style [6 Marks] → Think as an Engineer → New Context → Reflect & Challenge → Exercise.

Lessons covered

- 4-1 Types and Characteristics of Media
- 4-2 Information Design and User Experience for Websites
- 4-3 Methods for Evaluating Websites
- 4-4 The Iterative Improvement Process for Websites

Based on the supplied Egyptian Baccalaureate textbook, Part One.

Chapter 4 — Quick Revision

4-1 • MEDIA

TV = wide reach; radio = sound and can be received almost anywhere; newspaper = printed text/images; website = diverse and updatable; SNS = rapid sharing and two-way interaction.

4-2 • DESIGN

Persona = who the typical user is. Wireframe = where information goes and its priority. CRAP = Contrast, Repetition, Alignment, Proximity. UCD = design from the user's perspective.

4-3 • EVALUATION

Qualitative = why. Quantitative = how much. PV = page views; bounce rate = visits ending after the first page; CVR = visits reaching a target action. Heuristic evaluation = experts use rules of thumb.

4-4 • IMPROVEMENT

PDCA = Plan → Do → Check → Act, repeated. Design thinking = Empathize → Define → Ideate → Prototype → Test. A/B testing compares two options using real users and data.

Worked Example — True / False

1. A website can present diverse information such as text, images, and video in combination.

Answer: ■ True.

2. TV is a medium that allows two-way interaction with users.

Answer: × False — TV is one-way.

3. SNS is highly effective at spreading information rapidly.

Answer: ■ True.

4. A website normally can be viewed even without an Internet connection.

Answer: × False — it normally requires an Internet connection unless cached.

Worked Example — Matching

5. Match: (a) detailed information viewable at any time; (b) users spread information by word of mouth; (c) inform many people at once with video. Choices: A TV, B Website, C SNS.

Answer: (a) B Website; (b) C SNS; (c) A TV.

Try — Short Answer

6. Name two strengths of websites.

Answer: Any two: diverse information; updatable; accessible from almost anywhere; two-way interaction; browser use without a dedicated app.

7. Name two constraints of websites.

Answer: Any two: normally needs Internet; users must actively access information; loading/display speed may vary.

Try — Multiple Choice

8. Which is NOT an appropriate website characteristic? A. Updated at any time. B. Automatically reaches users without active access. C. Combines text, images and video. D. Forms/comments allow interaction.

Answer: B.

9. Choose A TV, B Website, C SNS, or D Email: (a) detailed product information at any time; (b) word-of-mouth spreading; (c) video to many people at once; (d) direct campaign to customers.

Answer: (a) B; (b) C; (c) A; (d) D.

Exam-Style [6 Marks]

10. Explain why an organization might use a website, TV and SNS together rather than only one to promote a new service. [6] Refer to reach, depth and word-of-mouth.

Answer: TV gives wide reach; the website gives detailed information at any time; SNS lets users share and spread the message.

Think as an Engineer — Investigate & Decide

11. A public library wants to promote a free reading program. Survey ten classmates/family members about printed posters, Facebook, WhatsApp and local radio. Consider a family with no home Internet, a student and the librarian. Recommend a media plan with two reasons.

Answer: Use survey evidence. Start from the audience hardest to reach and choose the medium that already reaches them. Combining media by purpose is often appropriate.

New Context

12. A hospital wants to announce new opening hours and let patients ask questions. Name one medium for rapid reach and one for two-way interaction, and explain why.

Answer: TV or SNS for rapid reach; a website with forms/comments or SNS for two-way interaction.

Reflect & Challenge

13. Which medium do you personally trust most for important news, and why?

Answer: Answers vary. Justify using the medium's characteristics and your reason for trusting it.

14. Plan a media mix for a school event on a budget that rules out TV. Name one audience the mix might still miss.

Answer: Example: website for detail, SNS for spreading, direct messages for specific groups. Identify an audience that may not use those channels.

Exercise

15. Fill the blanks: A website cannot be viewed without an (a) connection. Information does not reach users unless they actively (b) it. In actual information dissemination, (c) media are often combined.

Answer: (a) Internet; (b) access/view; (c) multiple.

16. Which statement correctly describes media? A Radio conveys video and audio. B Newspapers spread information rapidly. C SNS allows users to post and share. D TV allows two-way interaction.

Answer: C.

17. Which statement correctly describes choosing media? A Always use one. B A website is always sufficient. C Choose according to purpose and target audience. D SNS makes TV/websites unnecessary.

Answer: C.

Worked Example — True / False

1. A persona is a typical user image set up as a concrete person.

Answer: ■ True.

2. A wireframe is a final-version design including color and decoration.

Answer: × False — it is a rough blueprint before visual decoration.

3. Proximity means grouping related information close together.

Answer: ■ True.

4. UCD prioritizes the convenience of the creator.

Answer: × False — UCD designs from the user's perspective.

Worked Example — Matching CRAP

5. Match: (a) highlight important information; (b) repeat style for unity; (c) align positions for order; (d) group related information. Choices: A Contrast, B Repetition, C Alignment, D Proximity.

Answer: (a) A; (b) B; (c) C; (d) D.

Try — Key Terms

6. What is a typical user image set up as a concrete person called?

Answer: Persona.

7. What is the blueprint for placement and priority of information called?

Answer: Wireframe.

8. What are the four principles represented by the English initials C-R-A-P?

Answer: Contrast, Repetition, Alignment, Proximity.

9. What is the user-perspective design approach and its abbreviation?

Answer: User-Centered Design (UCD).

Try — Multiple Choice

10. Which is NOT appropriate persona information? A user's age/occupation; B user's purpose; C website server performance; D user's challenges.

Answer: C.

11. Match: large bold headings; group product name/price/photo; align left edges; standardize heading font/color. Choices A Contrast, B Repetition, C Alignment, D Proximity.

Answer: A; D; C; B respectively.

Exam-Style [6 Marks]

12. Explain why a designer creates a persona and wireframe before choosing colors and fonts. [6] Refer to user needs and stakeholder agreement.

Answer: The persona keeps design focused on a concrete user's needs. The wireframe fixes placement and priority before visual preferences, making discussion and agreement with stakeholders easier.

Think as an Engineer — Investigate & Decide

13. A community clinic needs a website for opening hours, services and directions. Ask 8–10 people what they would look for first, group answers into user types including a low-vision user or someone with no home Internet, prioritize four homepage blocks, and choose one CRAP principle.

Answer: Use evidence rather than taste. Start from the most important/hardest-to-reach need, then choose the CRAP principle that fixes the weakest point.

New Context

14. A school wants parents to find their child's grades quickly on a cluttered results page. Name a persona, one wireframe decision and one CRAP principle, and state what each improves.

Answer: Example: parent checking grades; place grades in a high-priority block; use contrast to highlight them or proximity to group related results.

Reflect & Challenge

15. On a confusing website, which CRAP principle seems most broken, and why?

Answer: Answers vary. Explain whether the issue is priority, consistency, order/alignment, or grouping.

16. A clinic can fund only one accessibility improvement. Choose one CRAP principle and say what decoration/feature you would give up and why.

Answer: Any justified choice is acceptable if it prioritizes the user's needs and explains the trade-off.

Exercise

17. Fill the blanks: First create a (a) representing a typical user. Next create a (b), a blueprint for placement and priority. Keep the (c) — contrast, repetition, alignment and proximity — in mind.

Answer: (a) Persona; (b) Wireframe; (c) CRAP principles.

18. Which is NOT part of CRAP? A Contrast B Encryption C Alignment D Proximity.

Answer: B.

19. Which best describes UCD? A Developer convenience B Visual design first C Begin from what the user needs D Use newest technologies.

Answer: C.

Worked Example — True / False

1. Qualitative evaluation uses numerical data to evaluate quantitatively.

Answer: × False — numerical data is quantitative; qualitative explores users' voices and behavior.

2. PV is the number of times a page has been viewed.

Answer: ■ True.

3. The proportion of visits achieving a target action is the bounce rate.

Answer: × False — it is conversion rate (CVR).

4. Heuristic evaluation uses experts who identify UI problems using rules of thumb.

Answer: ■ True.

Worked Example — Indicators

5. Match: (a) page views; (b) visits ending after the first page; (c) visits achieving a target action. Choices A PV, B Bounce rate, C Conversion rate.

Answer: (a) A; (b) B; (c) C.

Try — Short Answer

6. What is the method that evaluates qualitatively by observing user voices and behavior?

Answer: Qualitative evaluation.

7. What is the method that evaluates quantitatively using numerical data?

Answer: Quantitative evaluation.

8. What is the two-letter abbreviation for page views?

Answer: PV.

9. What is the expert method based on rules of thumb?

Answer: Heuristic evaluation.

Try — Classification

10. Which are quantitative evaluation methods? A Usability testing B Web analytics C Interviews D A/B testing.

Answer: B and D.

11. Which describes conversion rate? A Page views B Visits ending after first page C Visits achieving a target action such as purchase or registration D Viewed pages relative to total pages.

Answer: C.

Exam-Style [6 Marks]

12. Explain why a website team should use both qualitative and quantitative evaluation rather than only one. [6] Refer to what numbers show and what users' voices show.

Answer: Quantitative evaluation shows how much is happening; qualitative evaluation explores why. Combining numerical evidence with user voices/behavior and expert checks gives a clearer, more accurate picture.

Think as an Engineer — Investigate & Decide

13. An online bookstore has high PV but very low conversion rate. What might that mean? Which method would tell you why? Consider a customer hidden by raw PV data.

Answer: Many people arrive but few take the target action. Qualitative testing/interviews can reveal why. Small-screen or first-time users may have difficulties PV alone cannot show.

New Context

14. A school's new website gets many visits, but most people leave after the first page. Name one quantitative indicator, one qualitative method and one likely fix.

Answer: Bounce rate; usability testing or interviews; change the page based on the observed cause.

Reflect & Challenge

15. If you could see only one metric, would you choose PV, bounce rate or CVR, and why?

Answer: Answers vary. Choose according to the goal and explain what the metric tells you.

16. A page has a high bounce rate. Design a short qualitative test that reveals the cause and say what you would change first.

Answer: Observe users performing a task and ask what confused them or what they expected; change the element connected to the observed cause.

Exercise

17. Fill the blanks: (a) evaluation observes user voices and behavior; (b) evaluation uses numerical data; (c) shows page views; (d) shows visits ending after the first page; (e) shows visits achieving a target action.

Answer: (a) Qualitative; (b) Quantitative; (c) PV; (d) Bounce rate; (e) Conversion rate/CVR.

18. Classify: (a) observe users exploring a site; (b) tally access counts/page views; (c) interview users. A Qualitative, B Quantitative.

Answer: (a) A; (b) B; (c) A.

19. Which describes heuristic evaluation? A Users explore the site; B Numerical data analysis; C Experts identify UI problems using rules of thumb; D Compare two patterns with data.

Answer: C.

Worked Example — True / False

1. PDCA repeats Plan → Do → Check → Act.

Answer: ■ True.

2. Design thinking solves problems from the developer's perspective.

Answer: × False — it works from the user's perspective.

3. A/B testing should change only one element at a time.

Answer: ■ True.

4. PDCA is complete after one execution.

Answer: × False — it is repeated for continuous improvement.

Worked Example — Design Thinking Matching

5. Match: understand user challenges; clarify problem; generate ideas; create a prototype; have users try it.
Choices A Empathize, B Define, C Ideate, D Prototype, E Test.

Answer: A; B; C; D; E respectively.

Try — Key Terms

6. What improvement method repeats Plan → Do → Check → Act?

Answer: PDCA cycle.

7. What user-perspective approach discovers problems and generates solutions?

Answer: Design thinking.

8. What method prepares two patterns and compares them with data?

Answer: A/B testing.

9. Name the five design-thinking steps in order.

Answer: Empathize → Define → Ideate → Prototype → Test.

Try — Multiple Choice

10. Which is "Check" in PDCA? A Set goal; B Change homepage; C Measure bounce rate after the change; D Consider another improvement.

Answer: C.

11. Which is NOT appropriate A/B testing? A Prepare two patterns; B Change multiple elements at once; C Wait for sufficient data; D Show each to actual users.

Answer: B.

Exam-Style [6 Marks]

12. Explain why a website team improves in repeated PDCA cycles instead of trying to build a perfect site in one go. [6] Refer to measuring results and acting on them.

Answer: The first version is a best guess. Real users reveal what works and what does not. PDCA allows a team to plan, make, measure and act on evidence, producing gradual improvement through repetition.

Think as an Engineer — Investigate & Decide

13. A school club's event page gets few sign-ups. Lead one measured PDCA loop: choose a starting number, identify one obstacle, plan one change, do it, check again and decide the next step. Choose one element for an A/B test and explain why only that element changes.

Answer: Start with a measurable number, identify the biggest obstacle, make one focused change, measure again and use the result for the next cycle. One A/B element is changed so its effect can be identified.

New Context

14. An online shop finds its “buy” button is rarely clicked. Describe one A/B test and one PDCA loop, and name the number that would show the change worked.

Answer: Create two button versions changing only one element, show them to actual users and compare clicks/conversion. In PDCA set a goal, make the change, measure it and decide the next step.

Reflect & Challenge

15. Think of a website you use often. What one small change might help most, and how would you measure whether it worked?

Answer: Answers vary. State a specific change and a measurable result.

16. You may run one A/B test this month. Choose one element, predict the winning option and state the number that would prove you right or wrong.

Answer: Any justified test is acceptable; change one element only and identify a measurable result such as clicks or conversion rate.

Exercise

17. Fill the terms: (a) repeated Plan → Do → Check → Act; (b) user-perspective problem/solution approach; (c) two-option data comparison.

Answer: (a) PDCA cycle; (b) Design thinking; (c) A/B testing.

18. Match the five design-thinking descriptions: understand user; clarify problem; generate ideas; create prototype; users try it.

Answer: Empathize; Define; Ideate; Prototype; Test.

Chapter 4 — Final Exam Review

The four lesson chains

4-1 CHOOSE: purpose + audience → medium.
4-2 DESIGN: user → persona → wireframe → CRAP → UCD.
4-3 EVALUATE: quantitative = how much; qualitative = why.
4-4 IMPROVE: PDCA + design thinking + A/B testing → repeated improvement.

Rapid-Fire Questions

1. Name the five media compared in 4-1.

Answer: TV, radio, newspaper, website, SNS.

2. Name the four CRAP principles.

Answer: Contrast, Repetition, Alignment, Proximity.

3. What does UCD mean?

Answer: User-Centered Design.

4. Qualitative vs quantitative?

Answer: Qualitative explains why; quantitative measures how much using numerical data.

5. What do PV, bounce rate and CVR mean?

Answer: PV = page views; bounce rate = visits ending after the first page; CVR = visits reaching a target action.

6. What is heuristic evaluation?

Answer: Experts identify UI problems using rules of thumb.

7. What does PDCA stand for?

Answer: Plan, Do, Check, Act.

8. Name the five design-thinking steps.

Answer: Empathize, Define, Ideate, Prototype, Test.

9. Why change only one element in A/B testing?

Answer: So the cause of the observed effect can be identified.

10. Why is website improvement iterative?

Answer: The first version is a best guess; real users and measured results show what should change, so teams improve repeatedly.

6-Mark Answer Tip

Do not only list terms. Connect each key concept to the situation in the question and explain the reason. Example: TV → wide reach; website → detailed information; SNS → sharing/word of mouth.

Final Memory Map

CHOOSE → Media

DESIGN → Persona / Wireframe / CRAP / UCD

EVALUATE → Qualitative / Quantitative / PV / Bounce / CVR / Heuristics

IMPROVE → PDCA / Design Thinking / A/B Testing